

Datasheet

DW01HA

One Cell Lithium-ion/Polymer Battery Protection IC With Built-in MOSFET

FORTUNE'S
Properties
For Reference Only

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1. General Description

The DW01HA battery protection IC is designed to protect lithium-ion/polymer battery from damage or degrading the lifetime due to overcharge, overdischarge, and/or overcurrent for one-cell lithium-ion/polymer battery powered systems, such as cellular phones.

The ultra-small package and less required external components make it ideal to integrate the DW01HA into the limited space of battery pack. The accurate $\pm 25\text{mV}$ overcharging detection voltage ensures safe and full utilization charging. The very low standby current drains little current from the cell while in storage.

2. Features

- With built-in N-MOSFET of low turn-on resistance.
- Reduction in Board Size due to Miniature Package SOT-23-6, DFN-5.
- Protection IC :
 - Ultra-Low Quiescent Current at $2.5\mu\text{A}$ ($V_{\text{CC}}=3.9\text{V}$).
 - Ultra-Low Overdischarge Current at $1.8\mu\text{A}$ ($V_{\text{CC}}=2.0\text{V}$).
 - Overcharge Protection Voltage
 $4.28\text{V} \pm 25\text{mV}$
 - Overdischarge Protection Voltage
 $2.4\text{V} \pm 100\text{mV}$
 - Overcurrent Protection Voltage
 $150\text{mV} \pm 15\text{mV}$
 - Auto Recovery function
- MOSFET :
 - $R_{\text{DS(ON)}} < 55\text{m}\Omega$
($V_{\text{GS}} = 3.7\text{V}$, $I_{\text{D}} = 1\text{A}$)

3. Ordering Information

DW01HA-x

Serial code from S、D*

(S : SOT-23-6 Green-Package)

(D : DFN-5 Green-Package)

TEMPERATURE RANGE

$-40^{\circ}\text{C} \sim +85^{\circ}\text{C}$

4. Applications

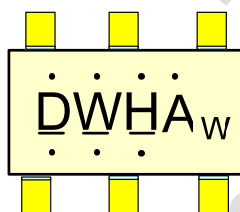
- Protection IC for One-Cell Lithium-Ion / Lithium-Polymer Battery Pack

5. Product Name List

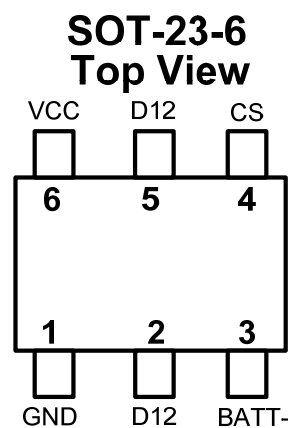
Model	Package	Overcharge detection voltage [VOC _P] (V)	Overcharge release voltage [VOCR] (V)	Overdischarge detection voltage [VOD _P] (V)	Overdischarge release voltage [VOD _R] (V)	Overcurrent detection voltage [VOI ₁] (V)	0V change function	Standby function release	Built-in N-MOSFET source to source Ron [RSS(ON)] (mΩ)
DW01HA	SOT-23-6	4.28±0.025	4.08±0.050	2.40±0.100	2.50±0.100	0.150±15mV	NO	AUTO recovery	< 55
DW01HA	DFN-5	4.28±0.025	4.08±0.050	2.40±0.100	2.50±0.100	0.150±15mV	NO	AUTO recovery	< 55

6. Pin Configuration and Package Marking Information

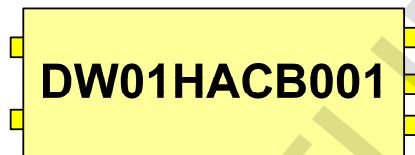
Pin No.	Symbol	Description
1	GND	Ground pin
2	D12	Two MOSFET common drain connection pin
3	BATT-	Connect to negative of charger or load
4	CS	Input pin for current sense, charger detect
5	D12	Two MOSFET common drain connection pin
6	VCC	Power supply, through a resistor (R1)



Top Point and Under_line : Lot No.
Bottom Point : Year
W : week, A~Z & A ~ Z



Pin No.	Symbol	Description
1	NC	NC
2	GND	Ground pin
3	BATT-	Connect to negative of charger or load
4	VCC	Power supply, through a resistor (R1)
5	CS	Input pin for current sense, charger detect
6	D12	Tow MOSFET common drain connection pin

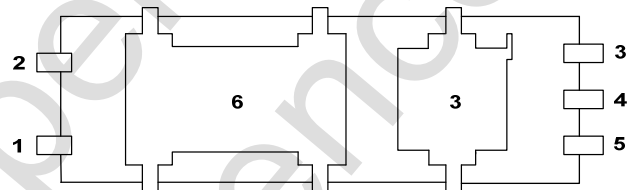


C : Year.
 B : Week Code, A~Z & A ~ Z
 001 :Serial number

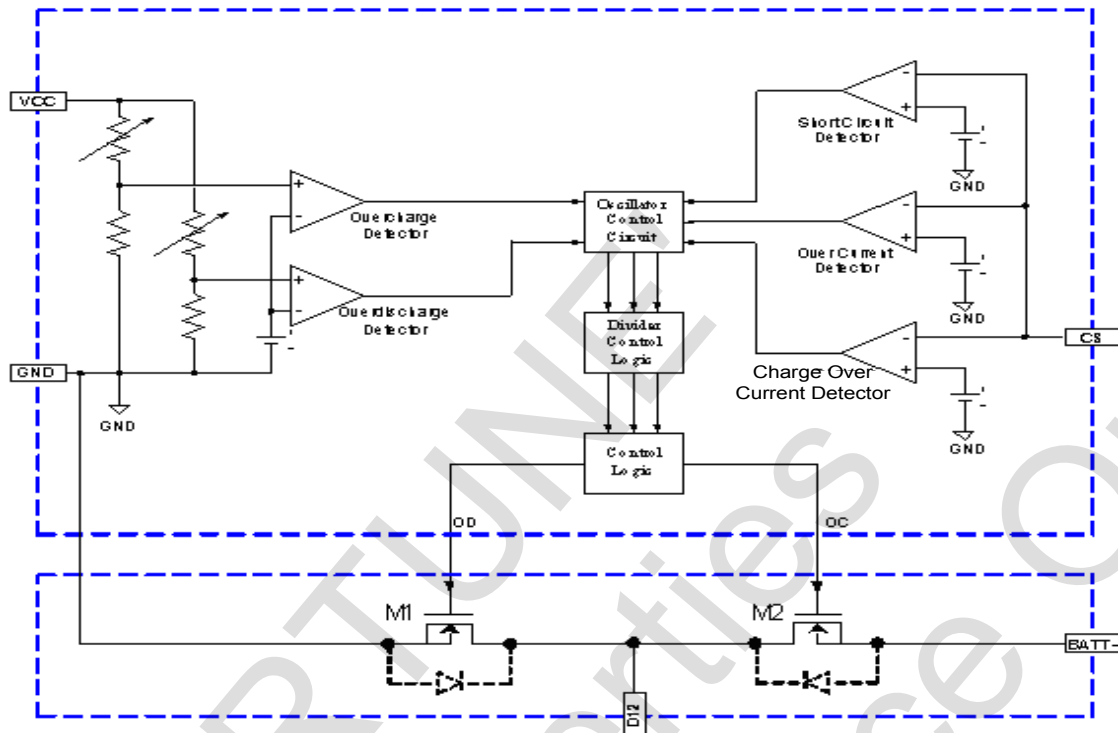
Top View



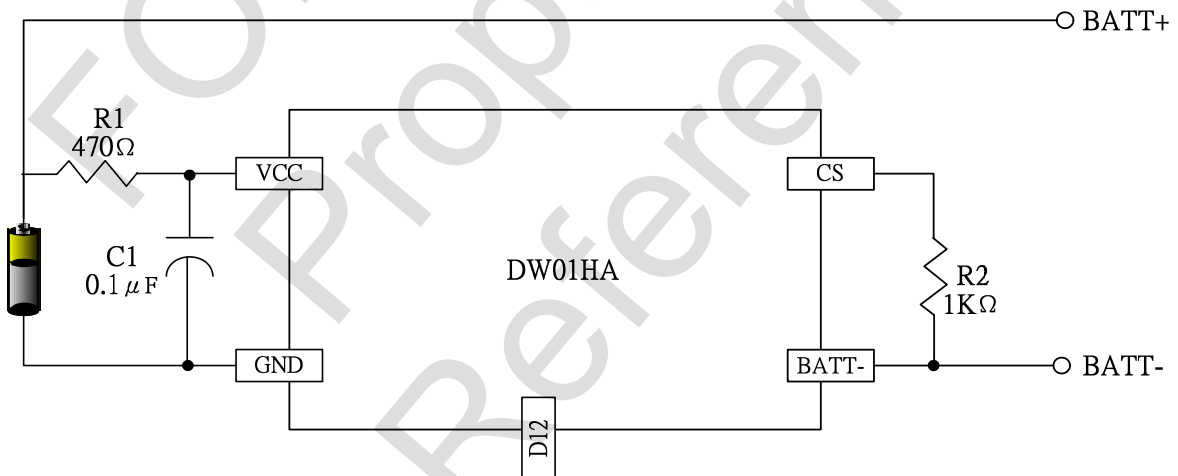
Bottom View



7. Functional Block Diagram



8. Typical Application Circuit



Symbol	Purpose	Recommended	Remakes
R1	ESD protection. For power fluctuation.	100~470Ω	Resistance should be as small as possible to avoid lowering of the overcharge detection accuracy caused by VDD pin current. Use 470Ω for better ESD protection.
C1	For power fluctuation.	0.1μF	
R2	Protection for reverse connection of a charger.	1k~2kΩ	Select a resistance as large as possible to prevent large current when a charge is connected in reverse.

9. Absolute Maximum Ratings

(GND=0V, Ta=25°C unless otherwise specified)

Item	Symbol	Rating	Unit
Input voltage between VCC and GND *	VCC	GND-0.3 to GND+15	V
CS input pin voltage	VCS	VCC -30 to VCC +0.3	V
Operating Temperature Range	TOP	-40 to +85	°C
Storage Temperature Range	TST	-40 to +125	°C
Drain-Source Voltage	VDS	20	V
Gate-Source Voltage	VGS	±12	V
Continuous Drain Current ³	ID @TA=25°C	6.8	A
Continuous Drain Current ³	ID @TA=100°C	4.3	A
Pulsed Drain Current ¹	IDM	25	A
Total Power Dissipation	PD @TA=25°C	1.4	W
Storage Temperature Range	TSTG	-55 to 150	°C
Operating Junction Temperature Rang	TJ	-55 to 150	°C

Note: DW01HA contains a circuit that will protect it from static discharge; but please take special care that no excessive static electricity or voltage which exceeds the limit of the protection circuit will be applied to it.

10. Electrical Characteristics

(Ta=25 °C unless otherwise specified)

PARAMETER	TEST CONDITIONS	SYMBOL	Min	Typ	Max	UNIT
Supply Current	VCC=3.9V	ICC		2.5	5.2	μA
Overdischarge Current	VCC=2.0V	IOD		1.8	4.5	μA
Overcharge Protection Voltage	DW01HA	VOCP	4.255	4.280	4.305	V
Overcharge Release Voltage		VOCR	4.030	4.080	4.130	V
Overdischarge Protection Voltage		VODP	2.300	2.400	2.500	V
Overdischarge Release Voltage		VODR	2.400	2.500	2.600	V
Overcurrent Protection Voltage		VOIP (VOI1)	0.135	0.150	0.165	V
Short Current Protection Voltage	VCC=3.6V	VSIP (VOI2)	0.60	0.70	0.80	V
Charger over current detection voltage	VCC=3.6V	VCOIP	-0.120	-0.100	-0.080	V
Faulty charger detect voltage		Vdet	6.0	8.0	10.0	V
Faulty charger recovery voltage		Vrec	5.8	7.3	8.8	V
0V charging prohibit		VST	0.65	0.95	1.25	V
Overcharge Delay Time		TOC	130	200	280	ms
Overdischarge Delay Time	VCC=3.6V to 2.0V	TOD	65	100	140	ms
Overcurrent Delay Time (1)	VCC=3.6V	TOI1	13.3	20.0	26.5	ms
Overcurrent Delay Time (2)	VCC=3.6V	TOI2	0.60	1.0	1.80	ms
Charge over current delay time	VCC=3.6V	Tdet	5.10	8.50	12.75	ms
N-MOSFET have low turn-on resistance						
Drain-Source Breakdown Voltage (BATT- to D12 / D12 to GND)	V _{GS} =0V, I _D =250μA	BV _{DSS}	20			V
Breakdown Voltage Temperature Coefficient	Reference to 25°C, I _D =1mA	$\Delta BV_{DSS} / \Delta T_j$		0.1		V/°C
Static Source-Source On-Resistance (BATT- to GND)	V _{GS} =3.7V, I _D =1A	R _{SS(ON)}		50	55	mΩ
Static Source-Source On-Resistance (BATT- to GND)	V _{GS} =2.7V, I _D =1A			60	70	mΩ
Drain-Source Leakage Current (BATT- to D12 / D12 to GND)	V _{DS} =16V, V _{GS} =0V	I _{DSS} (T _j =25°C)			1	μA
Drain-Source Leakage Current (BATT- to D12 / D12 to GND)	V _{DS} =16V, V _{GS} =0V	I _{DSS} (T _j =85°C)			30	μA

11. Description of Operation

Normal Condition

If $VODP < VCC < VOCP$ and $VCH < VCS < VOI1$, M1 and M2 are both turned on. The charging and discharging processes can be operated normally.

Overcharge Protection

When the voltage of the battery cell exceeds the overcharge protection voltage (VOCP) beyond the overcharge delay time (TOC) period, charging is inhibited by turning off of the charge control MOSFET. The overcharge condition is released in two cases:

The voltage of the battery cell becomes lower than the overcharge release voltage (VOCR) through self-discharge.

The voltage of the battery cell falls below the overcharge protection voltage (VOCP) and a load is connected.

When the battery voltage is above VOCP, the overcharge condition will not release even a load is connected to the pack.

Overdischarge Protection

When the voltage of the battery cell goes below the overdischarge protection voltage (VODP) beyond the overdischarge delay time (TOD) period, discharging is inhibited by turning off the discharge control MOSFET.

The default of overdischarge delay time is 100ms. Inhibition of discharging is immediately released when the voltage of the battery cell becomes higher than overdischarge release voltage (VODR) through charging.

Overcurrent Protection

In normal mode, the DW01HA continuously monitors the discharge current by sensing the voltage of CS pin. If the voltage of CS pin exceeds the overcurrent protection voltage (VOIP) beyond the overcurrent delay time (TOI1) period, the overcurrent protection circuit operates and discharging is inhibited by turning off the discharge control MOSFET. The overcurrent condition returns to the normal mode when the load is released or the impedance between BATT+ and BATT- is larger than 500kΩ. The DW01HA provides two overcurrent detection levels (0.15V and 0.7V) with two overcurrent delay time (TOI1 and TOI2) corresponding to each overcurrent detection level.

Charge Detection after Overdischarge

When overdischarge occurs, the discharge control MOSFET turns off and discharging is inhibited. However, charging is still permitted through the parasitic diode of MOSFET. Once the charger is connected to the battery pack, the DW01HA immediately turns on all the timing generation and detection circuitry. Charging progress is sensed if the voltage between CS and GND is below charge detection threshold voltage (VCH).

Auto Power Down recovery

The IC continues to operate even after the overdischarge state has been entered. The battery voltage rising to the overdischarge release voltage (VODR) or higher is the only required condition for the IC to return to the normal state.

12. Design Guide

Suppressing the Ripple and Disturbance from Charger

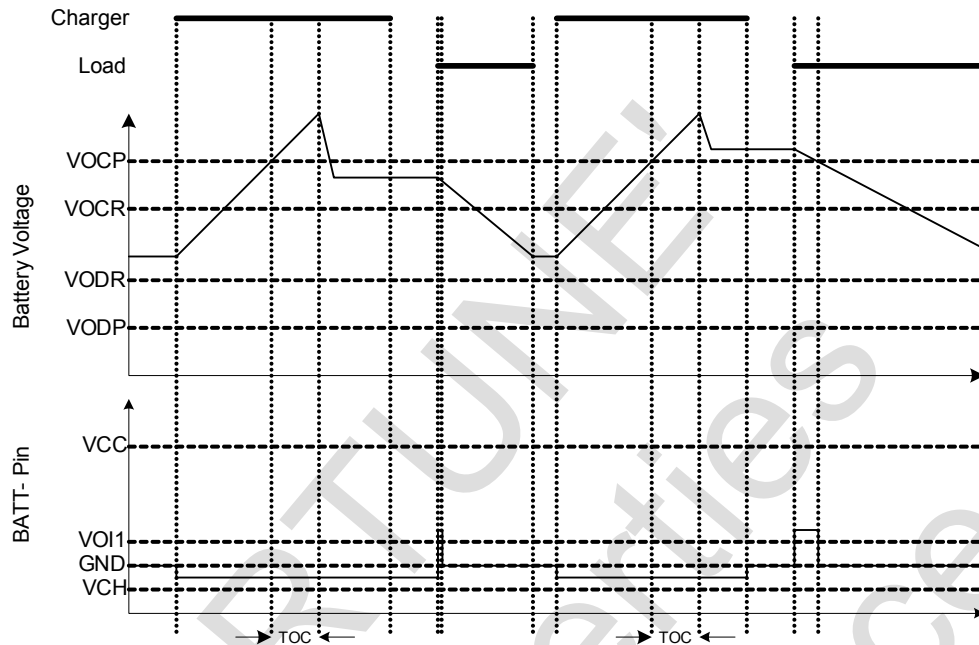
To suppress the ripple and disturbance from charger, connecting R1 and C1 to VCC is recommended.

Protection the CS pin

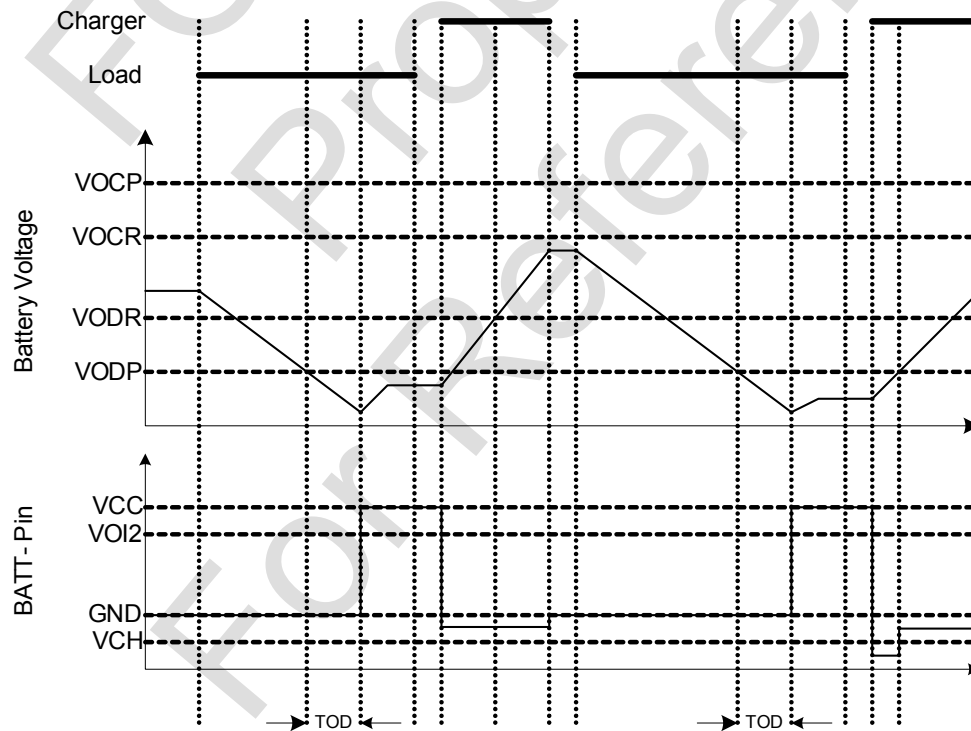
R2 is used for latch-up protection when charger is connected under overdischarge condition and overstress protection at reverse connecting of a charger.

13. Timing Diagram

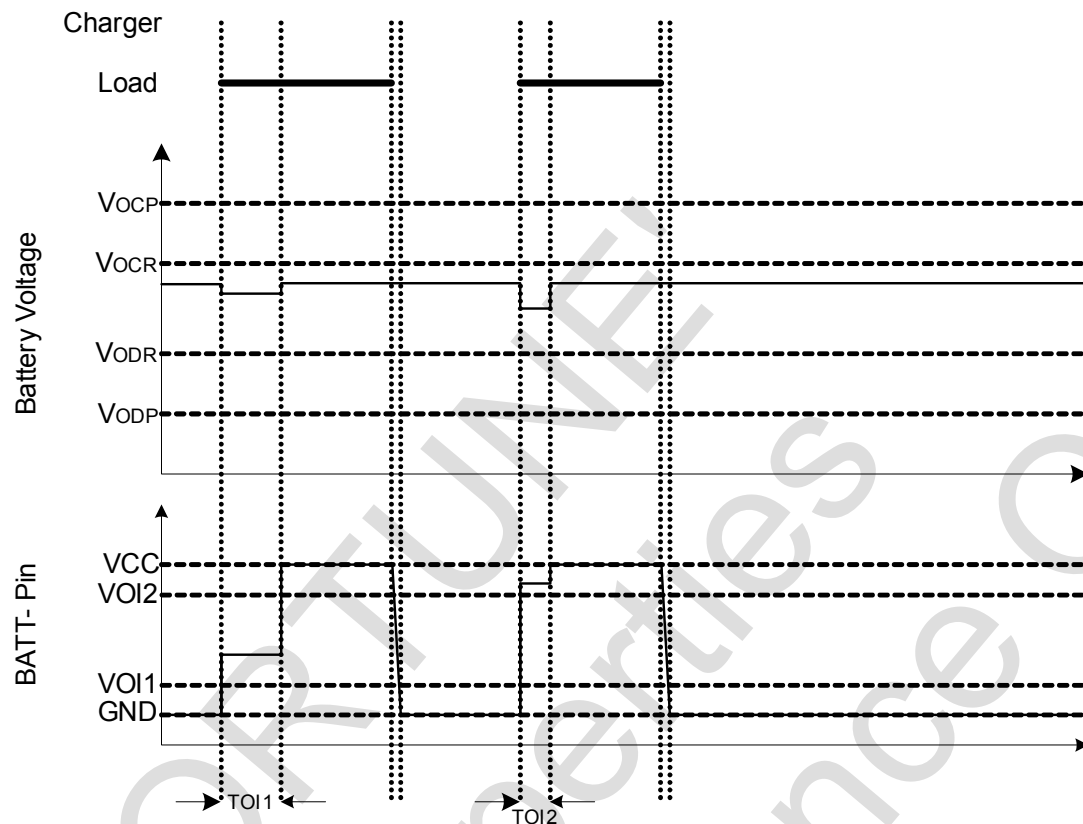
Overcharge Condition → Load Discharging → Normal Condition



Overdischarge Condition → Charging by a Charger → Normal Condition

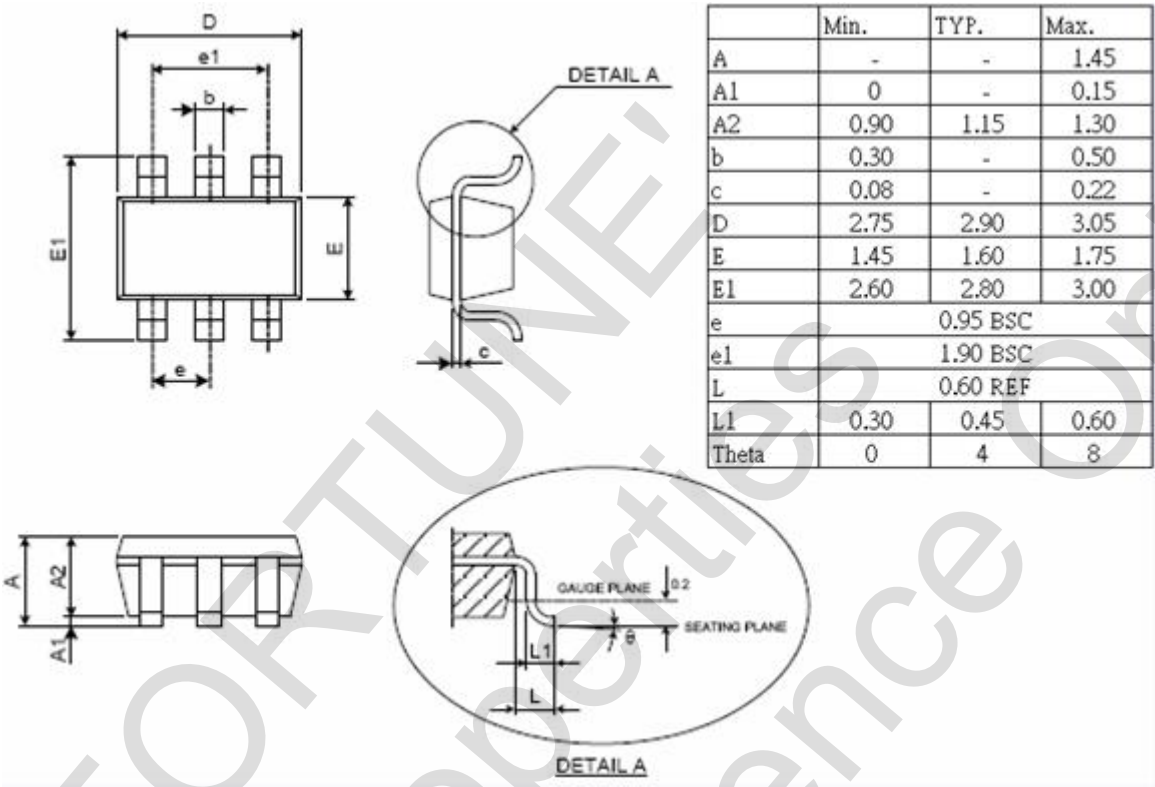


Over Current Condition → Normal Condition

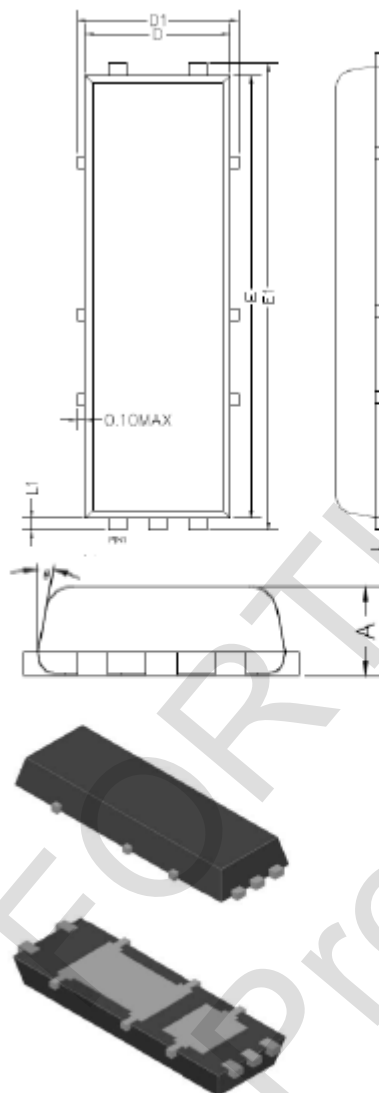


14. Package Outline

Dimension (SOT-23-6)



Dimension (DFN-5)



Unit: mm

Symbol	Min.	TYP.	Max.
A	0.58	0.63	0.68
A1	0.00	0.02	0.05
b	0.18	0.23	0.28
D	1.70	1.8	1.90
E	5.40	5.50	5.60
D1	-	-	2.10
E1	5.70	5.80	5.90
D2	1.10	1.20	1.30
E2	2.10	2.25	2.35
D3	1.10	1.20	1.30
E3	1.00	1.10	1.20
e	0.50 BSC		
e1	1.00 BSC		
K	0.67	0.77	0.87
K1	0.59	0.69	0.79
K2	0.25	0.35	0.45
L	0.25	0.35	0.45
L1	0.15 BSC		
θ	10°	12°	14°

Note:

All dimensions do not include mold flash, gate burrs or protrusions.

15. Revision History

Version	Date	Page	Description
1.0	2011/07/08	All	New release
1.1	2011/09/07	8, 13	Revise RSS(ON) 3.7V TYP : 50mΩ MAX : 55mΩ RSS(ON) 2.7V TYP : 60mΩ MAX : 70mΩ Revise Package Outline SOT-23-6
1.2	2012/09/12	14	Revise Package Outline (DFN-5)
1.3	2014/01/09	4	Revise VO11 Specified Unit